DAY 3

Linear Algebra

The Language of Machine Learning

Theory: 1 hour **Practical:** 2 hours

Topics: Scalars, Vectors, Matrices · Operations · Inverse · ML uses

LESSON OVERVIEW

What We'll Learn Today

1

0:00 – 1:00

Theory

Scalars, vectors, matrices, and operations

2

1:10 – 2:10

Practical 1

Compute by hand with small matrices

3

2:20 – 3:00

Practical 2

Pair practice + ML-style problems

💡 *Every ML model is built on vectors and matrices — let's see why!*

PAR

Theory • 1 hour

From single numbers to whole data tables

Why Linear Algebra for ML?

Linear algebra is how we put DATA and TRANSFORMATIONS into a computer.

Where you'll see it:

🖼️

Images

Each image is a matrix of pixel values

📊

Datasets

Rows = samples, columns = features

🧠

Neural nets

Layers are matrix multiplications

🔍

Search

Similarity = dot product of vectors

Scalars, Vectors, Matrices

Three levels of mathematical objects — each more structured than the last:

SCALAR

A single number

Example:

a = 5

Dimension: 0 dimensions

Notation: lowercase italic

VECTOR

A list of numbers

Example:

v = [3, 1, 4]

Dimension: 1 dimension (a line)

Notation: bold lowercase v

MATRIX

A grid of numbers

Example:

A = [[1, 2], [3, 4]]

Dimension: 2 dimensions (rows × cols)

Notation: BOLD UPPERCASE A

A Vector Is an Arrow

Think of a vector as an ARROW with:

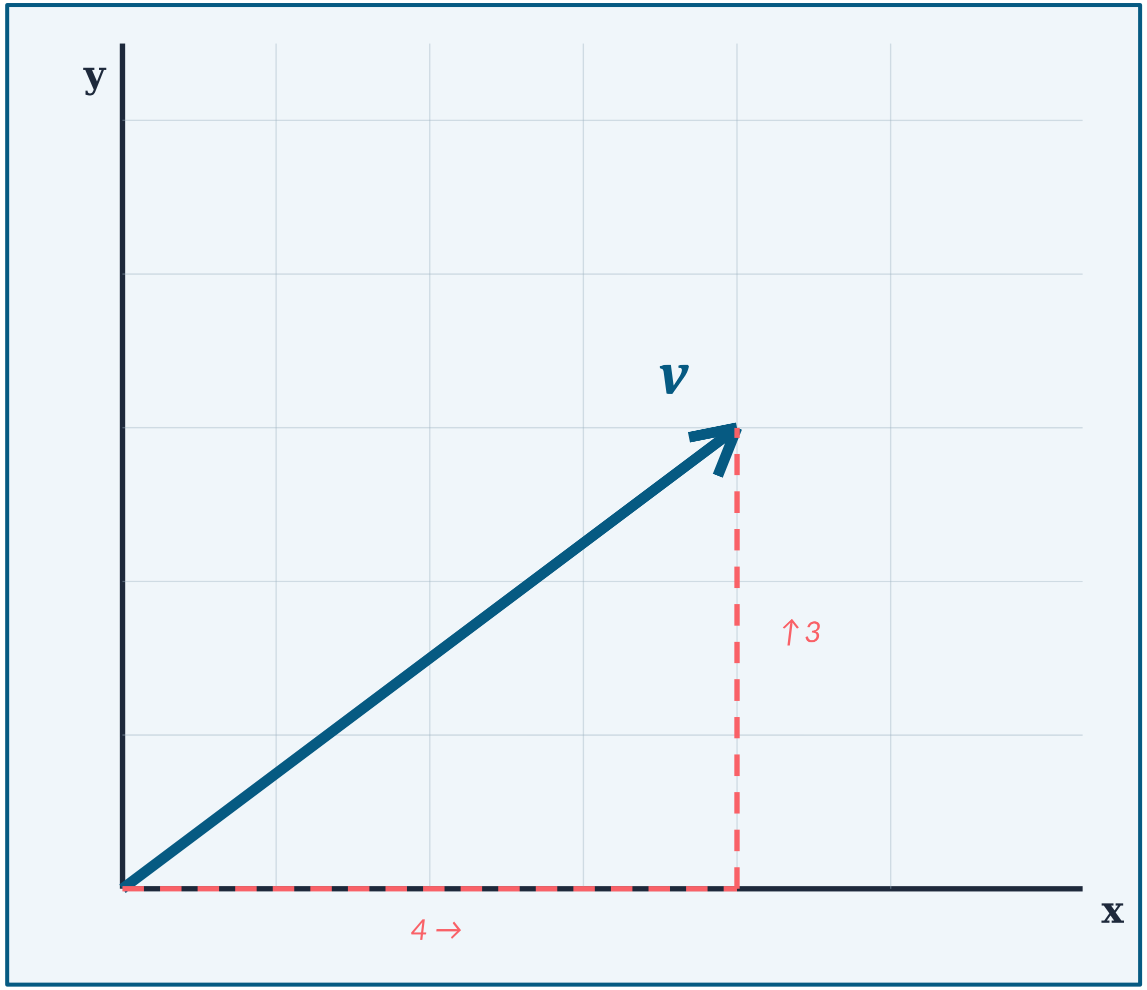
- **a direction**
- **a length (magnitude)**
- a starting point (usually the origin)

The vector v = [4, 3] is an arrow going:

4 steps RIGHT, then 3 steps UP

🔑 **In ML, a vector often represents a data point — each entry is a feature.**

e.g., a house: [bedrooms, bathrooms, square_feet, price] → [3, 2, 1400, 250000]



Addition & Scalar Multiplication

VECTOR ADDITION

$$u + v = [\, u_1+v_1,\ u_2+v_2,\ \ldots \,]$$

Add element-by-element.

Example:

$u = [1, 2]$, $v = [3, 4]$

$u + v = [1+3, 2+4] = [4, 6]$

🔑 Both vectors MUST have the same size.
You can't add [1,2] and [1,2,3].

SCALAR MULTIPLICATION

$$c \cdot v = [\, c{\cdot}v_1,\ c{\cdot}v_2,\ \ldots \,]$$

Multiply every entry by c (stretches / shrinks / flips the arrow).

Example:

$v = [2, 3]$, $c = 4$

$4 \cdot v = [4{\cdot}2, 4{\cdot}3] = [8, 12]$

🔑 $c > 1$ stretches, $0 < c < 1$ shrinks,
$c < 0$ flips the direction.

Turning Two Vectors into One Number

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u_1 \cdot v_1} + \mathbf{u_2 \cdot v_2} + \ldots + \mathbf{u_n \cdot v_n}$$

Recipe: multiply matching entries, then ADD them all up. Result is a SCALAR.

Worked Example

$\mathbf{u} = [1, 2, 3]$, $\mathbf{v} = [4, 5, 6]$. Find $\mathbf{u} \cdot \mathbf{v}$.

Step 1 — multiply matching entries: $1{\cdot}4 = 4$, $2{\cdot}5 = 10$, $3{\cdot}6 = 18$

Step 2 — add the products: $4 + 10 + 18$

Step 3 — result: $u \cdot v = 32$

🔑 **In ML, dot products measure SIMILARITY — used in search, recommendations, and neural nets.**

A Different Kind of Product

Don't worry — we won't compute it by hand today. Just get the idea.

📐

Works only in 3D

Dot product works for any size. Cross product ONLY for 3D vectors.

➡️

Result is a VECTOR

Unlike dot product (scalar), cross product gives back another vector.

⊥

Points PERPENDICULAR

The result is perpendicular to BOTH input vectors (at a right angle to both).

Everyday example: If your fingers curl from u to v, your thumb points along u × v.
Used in: physics (torque, magnetic force), 3D graphics (surface normals), game engines.

A Grid of Numbers

A matrix is a rectangular grid of numbers.

Size is $m \times n$ (rows × columns).

Notation:

- BOLD UPPERCASE letter: A, B, X
- Entry at row i, col j: a_{ij}
- Index starts from 1 (or 0 in code)

In the example →

A is 2 × 3 (2 rows, 3 columns)
$a_{11} = 1$, $a_{13} = 3$, $a_{22} = 5$

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

2 × 3

rows × columns

Addition & Transpose

MATRIX ADDITION

$$(A + B)_{ij} = A_{ij} + B_{ij}$$

Add matching entries — just like vectors.

Example:

A = [[1,2],[3,4]],   B = [[5,6],[7,8]]

A + B = [[6, 8], [10, 12]]

🔑 Matrices MUST be the same shape.
A 2×2 can't be added to a 3×3.

TRANSPOSE

$$A^{S} = \text{swap rows} \rightleftarrows \text{columns}$$

Flip the matrix across its diagonal.

Example:

A = [[1, 2, 3], [4, 5, 6]]   (2×3)

A^S = [[1, 4], [2, 5], [3, 6]]   (3×2)

🔑 Row 1 of A becomes Column 1 of A^T.
An m×n matrix becomes n×m.

The Most Important Operation

$(A \cdot B)_{ij}$ = row i of A · column j of B (a dot product!)

⚠️ **SIZE RULE:** A (m × k) times B (k × n) → result (m × n). The INNER numbers must match!

Worked Example: A = [[1, 2], [3, 4]] and B = [[5, 6], [7, 8]]. Find A · B.

Entry (1,1) — row 1 of A · col 1 of B: [1,2]·[5,7] = 1·5 + 2·7 = 5 + 14 = 19

Entry (1,2) — row 1 of A · col 2 of B: [1,2]·[6,8] = 1·6 + 2·8 = 6 + 16 = 22

Entry (2,1) — row 2 of A · col 1 of B: [3,4]·[5,7] = 3·5 + 4·7 = 15 + 28 = 43

Entry (2,2) — row 2 of A · col 2 of B: [3,4]·[6,8] = 3·6 + 4·8 = 18 + 32 = 50

Result: A · B = [[19, 22], [43, 50]]

How Entry (i, j) Gets Computed

For entry (1, 1): take ROW 1 of A (highlighted yellow) and COL 1 of B (highlighted red), dot-product them.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Entry (1, 1) = (yellow row) · (red column)

$= [1, 2] \cdot [5, 7] = 1{\cdot}5 + 2{\cdot}7 = 5 + 14 = 19$ ← goes in the GREEN box

Repeat for each entry of the result (row i of A, col j of B).

⚠️ WARNING: Matrix multiplication is NOT commutative. $A \cdot B \neq B \cdot A$ (in general).

The "1" of Matrices

Just like $1 \cdot x = x$ for numbers,

the IDENTITY MATRIX I does nothing.

$$I \cdot A = A = A \cdot I$$

Shape:

- Always square ($n \times n$).
- 1's on the diagonal, 0's everywhere else.
- Written as I_n or just I.

🔑 **Multiplying by I leaves the matrix UNCHANGED.**

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(diagonal = 1, off-diagonal = 0)

The "÷" of Matrices

$$A \cdot A^{-1} = I = A^{-1} \cdot A$$

Just like $5 \cdot (1/5) = 1$, the inverse A^{-1} "undoes" A.

1

Square only

Only square matrices ($n \times n$) can have an inverse.

2

Not always exists

Some matrices don't have an inverse (called SINGULAR).

3

Solve equations

If $A \cdot x = b$, then $x = A^{-1} \cdot b$. Exactly like dividing!

Linear Algebra in Machine Learning

📊 REPRESENTING DATA

Datasets are matrices.

	Bed	Bath	Sq.ft
H_1	3	2	1400
H_2	4	3	2100
H_3	2	1	900

Each ROW is a data point. Each COLUMN is a feature.

This whole table = one matrix X.

🔄 TRANSFORMATIONS

Matrices transform data.

Examples:

- Rotate an image → multiply pixels by a rotation matrix.
- Neural network layer → output = W · input + b
- Dimension reduction (PCA) → project onto smaller space.
- Compress images / text → low-rank matrix factorisation.

🔑 ML = data (matrices) × transformations (more matrices)

Theory — What We Learned

1

3 objects

Scalars (numbers), vectors (lists), matrices (grids)

2

Vector ops

Add, scalar mult., dot product (similarity)

3

Matrix ops

Add, multiply (row·col), transpose (flip)

4

Identity & inverse

I does nothing; A^{-1} undoes A

☕ ***Break time — see you in 10 minutes, then we PRACTICE!***

P A R

Practical • 2 hours

Compute with small matrices by hand

PRACTICAL · EXAMPLE 1

Worked Example (Together on the Board)

Question:

Compute $u + v$ where $u = [2, 3, 5]$ and $v = [1, 4, 2]$.

Solution:

Step 1: Check both vectors have the SAME size. u has 3 entries, v has 3 ✓

Step 2: Add matching entries: (2+1), (3+4), (5+2)

Step 3: Compute each: 3, 7, 7

Answer: $u + v = [3, 7, 7]$

PRACTICAL · EXAMPLE 2

Worked Example (Together on the Board)

Question:

Compute $3 \cdot v$ where $v = [2, -1, 4]$.

Solution:

Step 1:	Rule: multiply EACH entry by the scalar 3.
Step 2:	Multiply: $3{\cdot}2,\ 3{\cdot}(-1),\ 3{\cdot}4$
Step 3:	Simplify: $6,\ -3,\ 12$

Answer: $3 \cdot v = [6, -3, 12]$

PRACTICAL · EXAMPLE 3

Worked Example (Together on the Board)

Question:

Compute $u \cdot v$ where $u = [2, 3, 4]$ and $v = [1, 0, 2]$.

Solution:

Step 1: Same size? u has 3, v has 3 ✓

Step 2: Multiply matching entries: $2{\cdot}1 = 2$, $3{\cdot}0 = 0$, $4{\cdot}2 = 8$

Step 3: Sum them up: $2 + 0 + 8$

Step 4: Total = 10. Note: result is a single NUMBER (scalar).

Answer: $u \cdot v = 10$

Worked Example (Together on the Board)

Question:

Compute $A + B$ where $A = [[1, 2], [3, 4]]$ and $B = [[5, 1], [0, 2]]$.

Solution:

Step 1: Same shape? Both are 2×2 ✓

Step 2: Add entry (1,1): $1 + 5 = 6$

Step 3: Add entry (1,2): $2 + 1 = 3$

Step 4: Add entry (2,1): $3 + 0 = 3$

Step 5: Add entry (2,2): $4 + 2 = 6$

Answer: $A + B = [[6, 3], [3, 6]]$

PRACTICAL · EXAMPLE 5

Worked Example (Together on the Board)

Question:

Find A^S where $A = [[1, 2, 3], [4, 5, 6]]$.

Solution:

Step 1: A is 2×3. Transpose flips the shape → A^T will be 3×2.

Step 2: Row 1 of $A = [1, 2, 3]$ becomes Column 1 of A^T

Step 3: Row 2 of $A = [4, 5, 6]$ becomes Column 2 of A^T

Step 4: Stack them as columns: col1 = [1, 2, 3], col2 = [4, 5, 6]

Answer: $A^S = [[1, 4], [2, 5], [3, 6]]$

Worked Example (Together on the Board)

Question:

Compute $A \cdot B$ where $A = [[1, 2], [0, 1]]$ and $B = [[3, 0], [1, 4]]$.

Solution:

Step 1: Size check: A is 2×2, B is 2×2. Inner 2 = 2 ✓. Result will be 2×2.

Step 2: Entry (1,1) = row1(A) · col1(B) = [1,2]·[3,1] = 1·3 + 2·1 = 5

Step 3: Entry (1,2) = [1,2] · [0,4] = 1·0 + 2·4 = 8

Step 4: Entry (2,1) = [0,1] · [3,1] = 0·3 + 1·1 = 1

Step 5: Entry (2,2) = [0,1] · [0,4] = 0·0 + 1·4 = 4

Answer: $A \cdot B = [[5, 8], [1, 4]]$

Your Turn — Work in Pairs

👥 How this works

- Pair up with the person next to you.
- Work through the practice questions on the next slides.
- Explain your steps to your partner out loud.
- Stuck? Raise your hand — we'll help!
- After ~30 min, we'll review every answer together.

📝 ***Remember: ALWAYS check matrix sizes FIRST before computing!***

P R A C T I

10 Practice Questions

Solutions follow — try them FIRST, then check!

PRACTICE QUESTIONS

Questions 1 – 5

Q1

If $\mathbf{u} = [1, 4, 2]$ and $\mathbf{v} = [3, -1, 5]$, find $\mathbf{u} + \mathbf{v}$.

💡 *Hint: Add entry by entry*

Q2

Compute $5 \cdot \mathbf{v}$ where $\mathbf{v} = [2, 0, -3]$.

💡 *Hint: Scalar multiplication*

Q3

Find $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = [1, 2]$ and $\mathbf{v} = [4, 3]$.

💡 *Hint: Dot product → single number*

Q4

Add $A = [[2, 1], [0, 3]]$ and $B = [[1, 4], [5, 2]]$.

💡 *Hint: Entry-by-entry*

Q5

Find A^S for $A = [[1, 5], [2, 6], [3, 7]]$.

💡 *Hint: Rows ⇄columns*

PRACTICE QUESTIONS

Questions 6 – 10

Q6 **Compute $A \cdot B$ where $A = [[1, 0], [2, 1]]$ and $B = [[3, 4], [1, 2]]$.**

💡 *Hint: Row-col dot products*

Q7 **Is matrix multiplication commutative? (Does $A \cdot B$ always equal $B \cdot A$?)**

💡 *Hint: Try A=[[1,1],[0,1]], B=[[1,0],[1,1]]*

Q8 **What is $I_3 \cdot A$ if $A = [[5, 6], [7, 8], [9, 0]]$?**

💡 *Hint: I does nothing*

Q9 **Can you multiply a 3×2 matrix by a 2×4 matrix? If yes, what's the result's shape?**

💡 *Hint: Check inner numbers; result uses outer*

Q10 **A dataset has 100 houses, each with 4 features (bed, bath, sq.ft, price). What shape is the data matrix X?**

💡 *Hint: Rows = samples, cols = features*

S O L U

Answer Key

Check your work — and understand every step!

SOLUTION

Q1: Vector Addition

Problem: $u = [1, 4, 2]$, $v = [3, -1, 5]$. **Find** $u + v$.

Step-by-step:

- Check same size: u has 3, v has 3 ✓
- Add entries: $1 + 3 = 4$, $4 + (-1) = 3$, $2 + 5 = 7$
- Stack them back into a vector.

✓ Answer: $u + v = [4, 3, 7]$

SOLUTION

Q2: Scalar Multiplication

Problem: $5 \cdot v$ where $v = [2, 0, -3]$.

Step-by-step:

- Multiply EACH entry by 5:
- $5{\cdot}2 = 10, \quad 5{\cdot}0 = 0, \quad 5{\cdot}(-3) = -15$
- Put them back into a vector.

✓ **Answer:** $5 \cdot v = [10, 0, -15]$

SOLUTION

Q3: Dot Product

Problem: $\mathbf{u} = [1, 2]$, $\mathbf{v} = [4, 3]$. Find $\mathbf{u} \cdot \mathbf{v}$.

Step-by-step:

- Same size: both have 2 entries ✓
- Multiply matching entries: $1{\cdot}4 = 4$, $2{\cdot}3 = 6$
- Sum: $4 + 6 = 10$
- Result is a SCALAR (single number).

✓ Answer: $\mathbf{u} \cdot \mathbf{v} = 10$

SOLUTION

Q4: Matrix Addition

Problem: $A = [[2, 1], [0, 3]]$, $B = [[1, 4], [5, 2]]$.

Step-by-step:

- Both are 2×2 ✓ — shapes match.
- Entry (1,1): $2 + 1 = 3$ Entry (1,2): $1 + 4 = 5$
- Entry (2,1): $0 + 5 = 5$ Entry (2,2): $3 + 2 = 5$

✓ **Answer:** $A + B = [[3, 5], [5, 5]]$

SOLUTION

Q5: Transpose

Problem: $A = [[1, 5], [2, 6], [3, 7]]$. Find A^S.

Step-by-step:

- A is 3×2. Transpose flips shape → A^T is 2×3.
- Row 1 of $A = [1, 5]$ becomes Column 1 of A^T
- Row 2 of $A = [2, 6]$ becomes Column 2 of A^T
- Row 3 of $A = [3, 7]$ becomes Column 3 of A^T
- Stacking columns: first row = $[1, 2, 3]$, second row = $[5, 6, 7]$

✓ Answer: $A^S = [[1, 2, 3], [5, 6, 7]]$

SOLUTION

Q6: Matrix Multiplication

Problem: $A = [[1, 0], [2, 1]]$, $B = [[3, 4], [1, 2]]$. Find $A \cdot B$.

Step-by-step:

- Size check: $2{\times}2 \cdot 2{\times}2$ ✓. Result is $2{\times}2$.
- (1,1): $[1,0]\cdot[3,1] = 1\cdot3 + 0\cdot1 = 3$
- (1,2): $[1,0]\cdot[4,2] = 1\cdot4 + 0\cdot2 = 4$
- (2,1): $[2,1]\cdot[3,1] = 2\cdot3 + 1\cdot1 = 7$
- (2,2): $[2,1]\cdot[4,2] = 2\cdot4 + 1\cdot2 = 10$

✓ **Answer:** $A \cdot B = [[3, 4], [7, 10]]$

SOLUTION

Q7: Is MatMul Commutative?

Problem: Does $A \cdot B$ always equal $B \cdot A$?

Step-by-step:

- NO — in general matrix multiplication is NOT commutative.
- Test with $A = [[1, 1], [0, 1]]$, $B = [[1, 0], [1, 1]]$:
- $A \cdot B = [[2, 1], [1, 1]]$
- $B \cdot A = [[1, 1], [1, 2]]$
- They're different — so $A{\cdot}B \neq B{\cdot}A$ in this case.

✓ Answer: No — $A \cdot B \neq B \cdot A$ in general

SOLUTION

Q8: Identity Matrix

Problem: $I_3 \cdot A$ where $A = [[5, 6], [7, 8], [9, 0]]$.

Step-by-step:

- I is the "1" of matrices — multiplying by it changes nothing.
- Rule: $I \cdot A = A$ (and $A \cdot I = A$).
- So no computation needed.

✓ Answer: $I_3 \cdot A = A = [[5, 6], [7, 8], [9, 0]]$

SOLUTION

Q9: Shape Check

Problem: Can you multiply 3×2 by 2×4? What's the result's shape?

Step-by-step:

- Rule: $(m \times k) \cdot (k \times n) \rightarrow (m \times n)$. INNER numbers must match.
- Here: $3{\times}2 \cdot 2{\times}4$. Inner numbers: 2 and 2 ✓ — they match.
- Outer numbers give result shape: 3×4.

✓ Answer: Yes — result is 3 × 4

SOLUTION

Q10: Shape of Data Matrix

Problem: 100 houses, each with 4 features. What shape is the data matrix X?

Step-by-step:

- ▶ Convention: ROWS = samples (data points), COLUMNS = features.
- ▶ 100 samples → 100 rows.
- ▶ 4 features → 4 columns.
- ▶ So X is a 100×4 matrix.

✓ Answer: X is 100×4 (100 rows, 4 columns)

🎉

Great work today!

You now understand vectors, matrices, their operations, and why they power ML.

You've completed the 3-day math foundations for ML! 🎓

Keep going — you've got this! 👋